CMAN AI Analysis - CF09

ECE 410/510 Spring 2026 | Vanisri Kyatham

Task 1: Dominant Kernel Identification

The dominant kernel is the 3x3 CONV2D dot product, the inner loop of an im2col style convolution. It runs inside CONV2D-im2col and accounts for 80-6% of total software runtime. The hardware design replaces exactly this kernel with 4 parallel MAC processing elements.

Kernel Dimensions

Parameter	Value
Kernel Spatial Size	3x3 = 9 taps
I/P Channels	1
O/P Channels / PEs	4 (NUM_PE = 4)
I/P Feature Map	32x32 = 1024 pixels
O/P Feature Map	(32-3+1) x (32-3+1) = 30x30 = 900 o/p patches

Data Types

Data	Type	Size
Input Pixels	INT8 Signed	1 byte each
Weights	INT8 Signed	1 byte each
Accumulator	INT32 Signed	4 bytes each

The INT32 accumulator is needed because the maximum partial sum is 9 x 127 x 127 = 145,161, which overflows INT8 and INT16.

Task 2: FLOPs Count

Each MAC operation = 2 FLOPs

Each PE does 9 MACs = 18 FLOPs

4 PEs running in parallel = 4 x 18 = 72 FLOPs per patch

Across all 900 output patches for one full 32x32:

Total FLOPs per invocation = 64,800 FLOPs

Task 3: Bytes Transferred

Reuse Pattern: GEMM-style weight reuse across output patches.

Lower Bound (No Data Reuse)

Every time a weight or pixel is needed, it goes all the way to off-chip memory and fetches it even if it was just used one patch ago.

Pixels: 9 pixels x 1 byte x 900 patches = 8,100 bytes

Weights for all 4 PEs: 4 x 9 weights x 1 byte x 900 patches = 32,400 bytes

Bytes_lower = (9 + 36) x 900 = 45 x 900 = 40,500 bytes

Lower bound = 40,500 bytes. This gives the lowest possible AI.

Upper Bound (Perfect On-Chip Data Reuse for Weights)

Weights loaded once: 4 x 9 x 1 byte = 36 bytes

Pixels streamed across all patches: 9 x 900 = 8,100 bytes

Bytes_upper = 36 + 8,100 = 8,136 bytes

Upper bound = 8,136 bytes. This gives the highest possible AI.

Task 4: Arithmetic Intensity

AI = Total FLOPs / Total Bytes. From Task 2: FLOPs = 64,800

Bound	FLOPs	Bytes	AI (FLOP/byte)
Low (No reuse)	64,800	40,500	1.60
Upper (Weight reuse)	64,800	8,136	7.96

Sky130A Platform at 100 MHz

Peak compute = 4 PEs x 1 MAC/cycle x 100 MHz x 2 FLOPs/MAC = 800 MOPS

Peak BW = 330 MB/s (AXI4-Lite 32-bit bus)

Ridge Point = 800 / 330 = 2.42 FLOP/byte

AI = 1.60 is left of the ridge (2.42) — Memory bound with no reuse.

AI = 7.96 is right of the ridge (2.42) — Compute bound with weight reuse.

At AI = 1.60, attainable performance = 330 x 1.60 = 528 MOPS

At AI = 7.96, attainable performance = Min(330 x 7.96, 800) = 800 MOPS (Compute ceiling)

Task 5: Bottleneck Identification and Improvement

Bottleneck: Hardware Interface Bandwidth (AXI4-Lite)

One patch takes 120 clock cycles. Only 9 of those cycles are actual MAC computation, that is 7.5% utilisation. The remaining 111 cycles are AXI4-Lite bus overhead while the MAC units sit idle waiting for data. The design is not short on compute or on-chip memory, it is starved by the interface.

Single Highest-Leverage Change

Replace the per-patch sequential AXI writes with a 9-deep streaming pixel FIFO. This collapses pixel delivery from 111 bus overhead cycles down to 9 FIFO cycles per patch, cutting total patch latency from 120 cycles to 30 cycles and raising throughput from 60 MOPS to 240 MOPS, a 4x improvement with no changes to the MAC array.